

Technical Publication

Diesel Engine
12V 396 TE 94

Description and
Operation Manual
M011125/10E

DETROIT DIESEL



Printed in Germany

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Le manuel devra être observé en vue d'éviter des incidents ou des endommagements pendant le service. Aussi recommandons-nous à l'exploitant de le mettre à la disposition du personnel chargé de l'entretien et de la conduite.

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Il manuale va consultato per evitare anomalie o guasti durante il servizio, per cui va messo a disposizione dall'utente al personale addetto alla manutenzione e alla condotta.

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Para evitar falhas ou danos durante a operação, os dizeres do manual devem ser respeitados. Quem explora o equipamento economicamente consequentemente deve colocá-lo à disposição do respectivo pessoal da conservação, e à disposição dos operadores.

Salvo alterações.

Notes for the user

This Manual consists of 10 main sections, marked by tab-index sheets.

These sections are listed on this page.

Each Section has its own detailed Table of Contents.

The contents is subdivided into Groups and Subgroups.

The Groups in the Main Sections B and G correspond to the Groups in the Spare Parts Catalogue and other Product Support publications.

Using the Table of Contents to trace an entry

- ↓ Table of Contents (general)
- ↓ Contents
- ↓ Group / Subgroup
- ↓ Heading

Using the Alphabetical Index to trace an entry

- ↓ Term in Alphabetic Index
- ↓ Group / Subgroup
- ↓ Heading

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Engine summary, technical data

B Description

System and subassembly descriptions

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Task descriptions

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Schedule and maintenance task chart

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M011125/10E

**Description and
Operation Manual**

**12 V 396
TE94**



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GERMANY

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Diesel Engine**Group A
Product Summary**

The product summary gives details of the engine and contains technical data such as engine power, general engine specifications, engine weight, oil and coolant capacities, settings and operational data.

ENGINE LAYOUT, ENGINE POWERKey to Engine Model Designation 12 V 396 TE 94

12	=	number of cylinders
V	=	vee configuration
396	=	engine series - 100 times displacement of one cylinder in litres
T	=	exhaust turbocharging
E	=	charge air cooling in split-circuit coolant system
9	=	marine engine
4	=	design digit

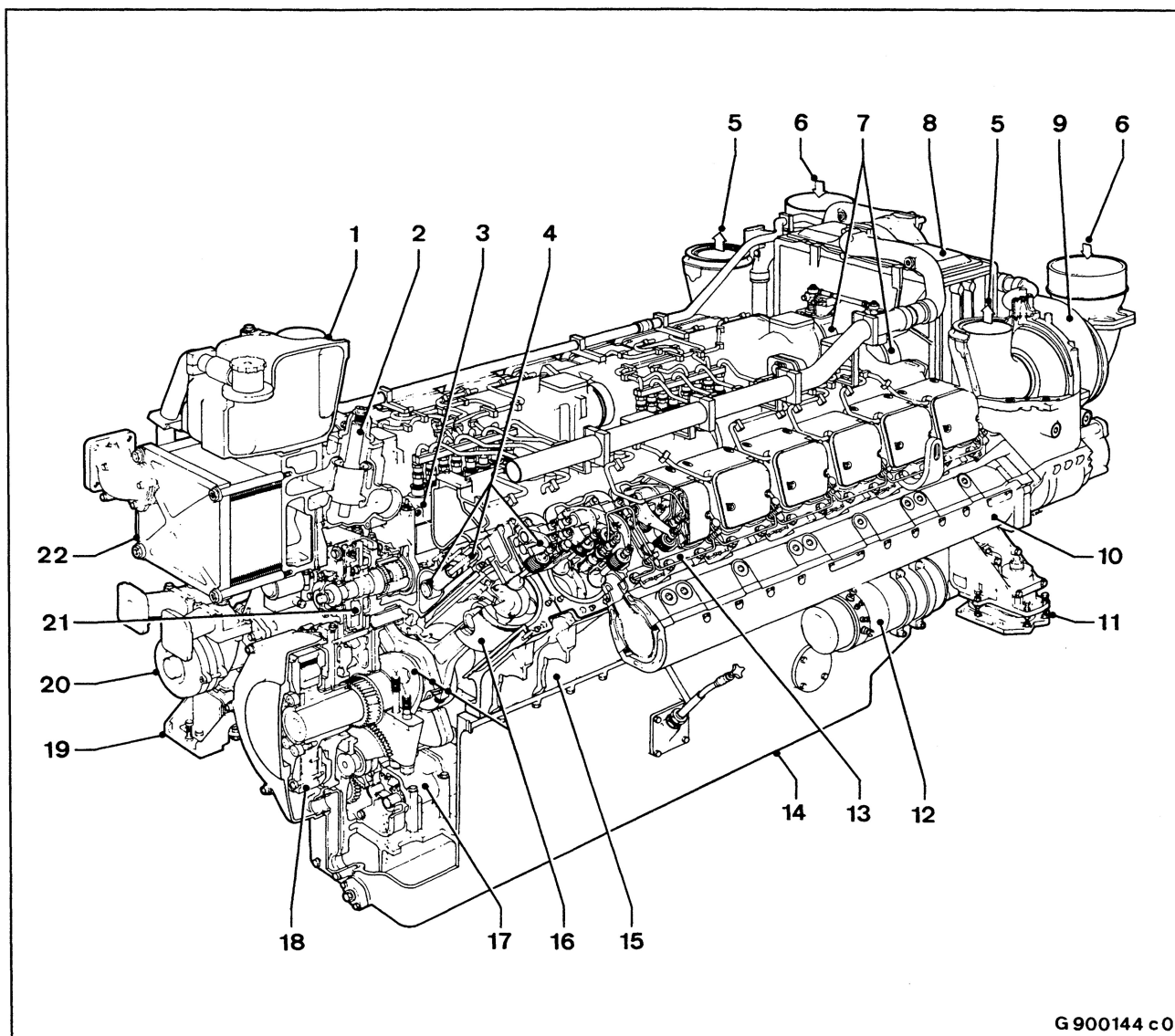
Engine Power

Fuel stop power 1680 kW at 2000 rpm to 2100 rpm

Reference conditions

- Intake air temperature	25 °C
- Raw water temperature	25 °C
- Fuel temperature	25 °C
- Intake pressure differential	20 mbar
- Exhaust pressure differential	30 mbar

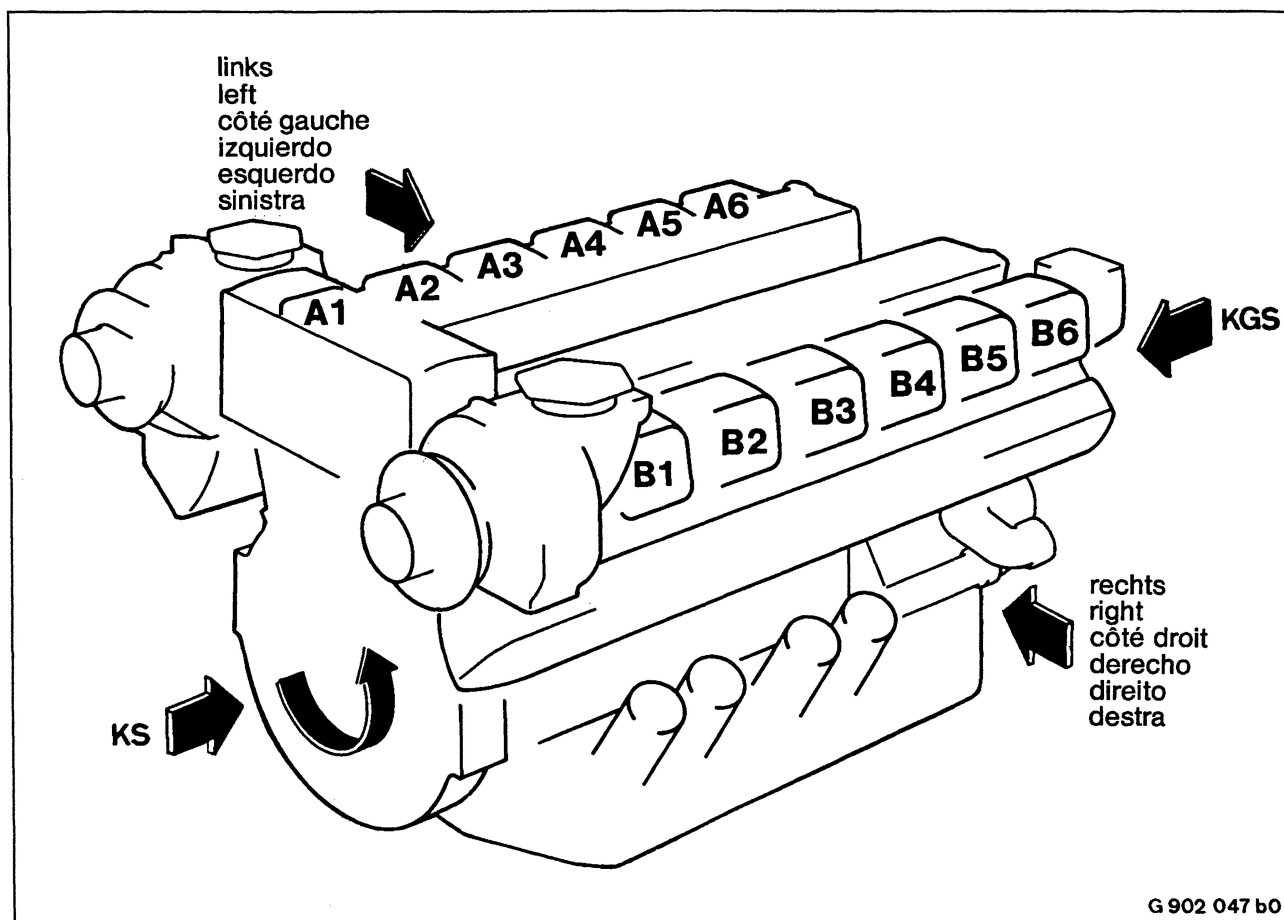
ENGINE LAYOUT, CUTAWAY DRAWING



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- | | |
|---------------------------------|---------------------------|
| 1 Expansion tank | 12 Starter |
| 2 Coolant thermostat | 13 Cylinder head |
| 3 Fuel injection pump | 14 Oil pan |
| 4 Valve gear | 15 Crankcase |
| 5 Exhaust outlet | 16 Running gear |
| 6 Air inlet | 17 Engine oil pump |
| 7 Emergency air shut-off flap | 18 Vibration damper |
| 8 Intercooler | 19 Engine mount, free end |
| 9 Exhaust turbocharger | 20 Bilge pump |
| 10 Exhaust manifold | 21 Injection timer |
| 11 Engine mounting, driving end | 22 Engine coolant cooler |

ENGINE SIDE AND CYLINDER DESIGNATIONS



Engine sides are always designated as viewed from the driving end.

The cylinders on the left engine side are designated "A" and those of the right engine side "B" (as per DIN ISO 1204).

The cylinders of each bank are numbered consecutively, starting with no. 1 at the driving end.

The numbering of other engine components is also from the driving end, starting with No. 1.

Designations and Abbreviations

Driving end	=	KS
Free end	=	KGS
Left side		
Right side		
Top		
Bottom		

GENERAL SPECIFICATIONS

Operating method	four-stroke cycle, single-acting
Combustion method	direct injection
Mode of supercharging	exhaust turbocharging single-stage sequential turbocharging
Cooling method	liquid-cooled
Cylinder arrangement	90° Vee
Bore	165 mm
Piston stroke	185 mm
Displacement, cylinder	3.96 l
Number of cylinders	12
Displacement, total	47.5 l
Compression ratio	12.3 : 1
Direction of rotation, viewed from driving end	c.c.w. (non reversible)
Firing order	A1 - B2 - A5 - B4 - A3 - B1 - A6 - B5 - A2 - B3 - A4 - B6
Firing order, cylinder cutout	A1 - A5 - A3 - A6 - A2 - A4
Nozzle opening pressure:	
Setting pressure with new spring	280 bar to 288 bar
Specified pressure with run-in spring	270 bar to 278 bar
Minimum pressure with run-in spring	260 bar
Final compression pressure at 120 rpm, 1000 mbar and engine coolant temperature of 5 °C	20 to 24 bar
Breakaway torque at 40 °C engine coolant temperature, allowance for acceleration included	approx. 600 Nm
Cranking torque at engine coolant temperature of 40 °C	approx. 560 Nm
Firing speed at engine coolant temperature of 40 °C	approx. 120 rpm
Mean piston speed	12.33 m/s at 2000 rpm
Engine noise, mean free-field level	approx. 106 dB(A) 1 m distance
Exhaust noise, undamped free-field level	approx. 119 dB(A) 1 m from outlet